Notes: Feenstra and Hanson (QJE, 1999)

The Impact of Outsourcing and High-Technology Capital on Wages: Estimates for the United States, 1979–1990

Contents

1 Big picture **2**

2 Incremental contribution of the paper **2**

3 Economic setting and intuition **3**

3.1 The empirical puzzle . . . 3

3.2 Why outsourcing can look like technological change . . . 3

4 The conventional price regression and why it becomes an identity **3**

4.1 Zero-profit logic . . . 3

4.2 The wage-differential term . . . 4

5 Two-stage decomposition methodology **4**

5.1 Stage 1: decompose price and productivity changes . . . 4

5.2 Stage 2: map each structural component into factor-price changes . . . 5

5.3 Interaction specification . . . 5

6 Factor bias versus sector bias **5**

7 Data and measurement **6**

7.1 Sample . . . 6

7.2 Outsourcing measures . . . 6

7.3 High-technology capital measures . . . 6

8 Main empirical results **7**

8.1 Price regression results . . . 7

8.2 Preliminary labor-demand regressions . . . 7

8.3 Linear two-stage decomposition . . . 7

8.4 Interaction decomposition . . . 7

8.5 Sensitivity to measuring computer capital . . . 8

9 How to read the paper's identification strategy **8**

10 Interpretation and economic message **8**

11 Tables and Figures **9**

11.1 Table I: Price Regression, 1979–1990; Table II: Summary Statistics 9

11.2 Figure I: Hicks-Neutral Technological Progress; Figure II: Skill-Biased Technological Progress 9

11.3 Table III: Nonproduction Wage Share; Table IV: Value-Added Prices plus Effective TFP 10

11.4 Table V: Estimated Factor-Price Changes; Table VI: Wage Inequality with Alternative High-Tech Measures 11

11.5 Table VII: Interaction First Stage; Table VIII: Interaction Wage-Inequality Decomposition 12

12 Short summary **13**

13 Conclusion **13**

1 Big picture

One-sentence summary. This note summarizes Feenstra and Hanson (1999), which develops a two-stage modification of the conventional price regression to decompose the rise in U.S. manufacturing wage inequality into components attributable to foreign outsourcing and high-technology capital, finding that both trade and technology matter, with computers usually explaining more of the increase in the nonproduction-production wage gap.

Core question. During the 1980s, why did the wage of more skilled workers rise relative to less skilled workers in U.S. manufacturing? The paper asks how much of the increase can be attributed to foreign outsourcing of intermediate inputs and how much can be attributed to high-technology capital, especially computers.

Key answer. In the base linear decomposition, computers explain about 35 percent of the observed increase in the relative wage of nonproduction workers, while outsourcing explains about 15 percent. In the richer interaction specification, narrow outsourcing explains about 40 percent and computer expenditures explain about 75 percent, although the exact quantitative allocation is sensitive to how computer capital is measured.

Why this paper matters. The paper is not only an empirical contribution to the trade-versus-technology debate. It is also a methodological paper: it shows that the conventional price regression is close to an accounting identity when total factor productivity and interindustry factor-price changes are fully included, so one must first explain price and productivity changes using structural variables before translating those changes into factor-price effects.

2 Incremental contribution of the paper

Relative to the trade-and-wages literature. Earlier papers often treated trade effects through product-price changes or factor-content calculations. Feenstra and Hanson instead measure trade as *foreign outsourcing of intermediate inputs*, which is closer to the way globalization changes production inside industries.

Relative to skill-biased technical change papers. Earlier work emphasized computers and other high-technology capital as drivers of skilled labor demand. This paper puts computers and outsourcing into the same empirical framework and estimates their relative contributions.

Relative to conventional price regressions. The paper shows that the usual price regression, if fully specified with effective TFP, becomes an identity and therefore cannot by itself identify the causal sources of wage changes. The contribution is to reinterpret the price regression as a second-stage decomposition tool after first decomposing price and productivity changes into structural components.

Relative to the factor-bias versus sector-bias debate. The methodology helps distinguish whether changes in product prices reflect neutral sectoral productivity shifts or nonneutral factor-biased technological change. If productivity pass-through to prices is complete and structural variables still directly affect prices, this is evidence of nonneutral technological change.

Empirical contribution. The paper provides estimates for U.S. manufacturing industries over 1979–1990 using detailed data on imported intermediate inputs, production/nonproduction wages, factor cost shares, and high-technology capital. It shows that the measured importance of computers

is highly sensitive to whether computer capital is measured using ex post rental prices, ex ante rental prices, or Census computer investment shares.

3 Economic setting and intuition

3.1 The empirical puzzle

The central empirical fact is the rise in the wage of nonproduction workers relative to production workers in U.S. manufacturing over 1979–1990. In the paper's data, the annual log wage change is 5.437 percent for nonproduction labor and 4.714 percent for production labor, so the relative nonproduction wage rises by roughly 0.72 percentage points per year.

The two candidate mechanisms are:

- **Foreign outsourcing:** U.S. industries import more intermediate inputs, especially production stages that are relatively intensive in less-skilled labor. This can raise the relative demand for skilled labor at home.
- **High-technology capital:** industries invest more in computers and related equipment, which may be complementary with skilled or nonproduction labor.

The difficulty is that both mechanisms can look similar in aggregate industry data: each can change industry productivity, product prices, and factor intensities. The paper's main task is to map these structural changes into implied changes in factor prices.

3.2 Why outsourcing can look like technological change

Outsourcing is treated as a change in the production technique. When an industry moves some production stages abroad, the set of activities remaining at home changes. If the outsourced activities are relatively production-labor intensive, the domestic activity bundle becomes more intensive in nonproduction labor. At the industry level, this can appear as a shift in total factor productivity and factor demand.

This is why foreign outsourcing can mimic skill-biased technical change in aggregate data: both can increase the relative demand for skilled or nonproduction labor without necessarily appearing as a simple change in final-good trade volumes.

4 The conventional price regression and why it becomes an identity

4.1 Zero-profit logic

The starting point is the zero-profit condition for industry i. Let p^v_{it} denote the value-added price, TFP_{it} total factor productivity, w_{it} the vector of primary factor prices, and s_{it} the vector of primary factor cost shares. A first-difference version of the zero-profit condition is

$$\Delta \ln p^v_{it} = -TFP_{it} + \frac{1}{2}(s_{i,t-1} + s_{it})' \Delta \ln w_{it}. \tag{1}$$

The conventional price regression replaces the industry-specific factor-price changes $\Delta \ln w_{it}$ with average manufacturing-wide factor-price changes ω_t and estimates

$$\Delta \ln p_{it}^v = -TFP_{it} + \frac{1}{2}(s_{i,t-1} + s_{it})'\omega_t + e_{it}. \tag{2}$$

The coefficients on factor shares are then interpreted as "mandated" or implied changes in factor prices.

4.2 The wage-differential term

The missing term is the deviation of each industry's factor-price change from the manufacturing-wide average:

$$e_{it} = \frac{1}{2}(s_{i,t-1} + s_{it})' \left(\Delta \ln w_{it} - \omega_t\right). \tag{3}$$

Feenstra and Hanson interpret this as a change in interindustry wage differentials, or more broadly as the effect of industry-specific factor-price deviations due to factor quality or rents.

They define effective TFP as

$$ETFP_{it} = TFP_{it} - e_{it}. \tag{4}$$

Substituting this definition into the price regression gives

$$\Delta \ln p_{it}^v = -ETFP_{it} + \frac{1}{2}(s_{i,t-1} + s_{it})'\omega_t. \tag{5}$$

Key implication. Once effective TFP is included, this equation fits almost exactly as an identity. Therefore, a conventional price regression cannot, by itself, explain the source of wage changes; it only restates the relation between prices, productivity, and factor prices.

5 Two-stage decomposition methodology

5.1 Stage 1: decompose price and productivity changes

The paper introduces a vector of structural variables z_{it}, including foreign outsourcing and high-technology capital. First, effective productivity is modeled as

$$ETFP_{it} = \alpha' \Delta z_{it} + \varepsilon_{it}. \tag{6}$$

Then value-added price changes are allowed to respond both to productivity and directly to structural variables:

$$\Delta \ln p_{it}^v = \lambda ETFP_{it} + \beta' \Delta z_{it} + \nu_{it}. \tag{7}$$

Combining the two gives the first-stage reduced form:

$$\Delta \ln p_{it}^v + ETFP_{it} = \gamma' \Delta z_{it} + \eta_{it}, \qquad \gamma = (1+\lambda)\alpha + \beta. \tag{8}$$

This first stage asks: how much of the total movement in value-added prices plus effective productivity is associated with each structural variable?

5.2 Stage 2: map each structural component into factor-price changes

For each structural variable k, the paper constructs the component

$$\gamma_k \Delta z_{itk}. \tag{9}$$

It then runs the second-stage modified price regression

$$\gamma_k \Delta z_{itk} = \delta_k' \frac{(s_{i,t-1} + s_{it})}{2} + u_{itk}. \tag{10}$$

The estimated vector δ_k is interpreted as the factor-price change attributable to structural variable k. This is the key decomposition step: instead of using all price and productivity changes, the paper uses only the part associated with outsourcing, computers, or other high-tech capital.

5.3 Interaction specification

The paper also estimates a richer model in which structural variables interact with primary factor quantities:

$$\Delta \ln p_{it}^v + ETFP_{it} = \gamma' \Delta z_{it} + \frac{1}{2} \Delta z_{it}' A' (\ln x_{i,t-1} + \ln x_{it}) + \epsilon_{it}. \tag{11}$$

For structural variable k, the corresponding component is

$$\gamma_k \Delta z_{itk} + \frac{1}{2} \Delta z_{itk} A_k' (\ln x_{i,t-1} + \ln x_{it}). \tag{12}$$

The second-stage regression becomes

$$\gamma_k \Delta z_{itk} + \frac{1}{2} \Delta z_{itk} A_k' (\ln x_{i,t-1} + \ln x_{it}) = \delta_k' \frac{(s_{i,t-1} + s_{it})}{2} + u_{itk}. \tag{13}$$

The interaction terms are intended to capture nonneutral technological change more directly. For example, a positive interaction between computer capital and nonproduction labor suggests complementarity between computers and skilled labor.

6 Factor bias versus sector bias

The paper clarifies an important conceptual distinction.

Sector-biased or Hicks-neutral productivity change. A neutral productivity increase in one sector shifts that sector's unit-cost curve outward. With Cobb-Douglas preferences and full pass-through, the resulting price decline can exactly offset the productivity gain, leaving factor prices unchanged.

Factor-biased technological change. A skill-biased productivity change shifts the unit-cost curve in a way that changes the relative demand for skilled and unskilled labor. Even if prices adjust, factor markets are not restored to their original equilibrium. Relative factor prices must change.

The figures in the paper illustrate this distinction. Figure I shows that Hicks-neutral progress in one industry can be neutralized through price adjustment. Figure II shows that skill-biased progress creates an additional feedback on goods prices and factor prices, so structural variables should enter directly in the price equation.

7 Data and measurement

7.1 Sample

The empirical sample is U.S. manufacturing from 1979 to 1990. The main industry data come from the NBER Productivity Database. The authors work with 447 four-digit SIC manufacturing industries after excluding three industries with missing materials data.

The key labor categories are:

- **Production workers:** used as the proxy for less-skilled labor.
- **Nonproduction workers:** used as the proxy for more-skilled labor.

7.2 Outsourcing measures

Imported intermediate inputs are constructed as

$$\sum_j [\text{input purchases of good } j \text{ by industry } i] \left[\frac{\text{imports of good } j}{\text{consumption of good } j}\right]. \tag{14}$$

This measure is divided by total expenditure on nonenergy intermediates.

The paper uses two measures:

- **Broad outsourcing:** imported intermediate inputs from all manufacturing industries.
- **Narrow outsourcing:** imported intermediate inputs from the same two-digit SIC industry as the buying industry.

The narrow measure is the preferred measure because it captures offshoring of production activities that could plausibly have been performed within the same industry.

7.3 High-technology capital measures

The paper uses several measures of high-technology capital:

- **BLS high-tech capital share:** includes office, computing, accounting machinery, communications equipment, scientific instruments, photocopying, and related equipment.
- **Computer share:** a narrower measure based on office, computing, and accounting machinery.
- **Ex post rental prices:** asset rental prices based on internal rates of return and capital gains.
- **Ex ante rental prices:** asset rental prices based on a safe rate of return and excluding capital gains.

- **Census computer investment share:** fraction of investment devoted to computer purchases.

A key empirical lesson is that the estimated contribution of computers is sensitive to this measurement choice.

8 Main empirical results

8.1 Price regression results

Table I demonstrates the methodological point. When effective TFP is included, the price regression nearly replicates the actual factor-price changes. When effective TFP is replaced by a less complete TFP measure or dropped, the estimated factor-price changes become highly sensitive to specification choices.

This is why the authors do not treat the conventional price regression as an independent causal model. Instead, they use it as a second-stage decomposition tool after assigning price and productivity changes to structural sources.

8.2 Preliminary labor-demand regressions

Table III estimates changes in the nonproduction wage share directly. It finds that outsourcing and computers are positively associated with the shift toward nonproduction labor. In these regressions, total outsourcing explains roughly 13–23 percent of the rise in the nonproduction wage share. Computer-related measures explain between about 8 percent and 32 percent, depending on how computer capital is measured.

8.3 Linear two-stage decomposition

In the first-stage linear decomposition, Table IV shows that narrow outsourcing and computers positively affect value-added prices plus effective TFP. Table V maps these components into factor-price changes.

The key base estimates are:

- Narrow outsourcing raises the relative nonproduction wage by about 0.108 percentage points per year.
- Computers raise the relative nonproduction wage by about 0.252 percentage points per year.
- Since the actual relative nonproduction wage rose by roughly 0.72 percentage points per year, narrow outsourcing explains about 15 percent and computers explain about 35 percent in the base specification.

8.4 Interaction decomposition

The interaction specification allows the impact of outsourcing and high-tech capital to depend on factor quantities. In Table VII, the interaction of narrow outsourcing with nonproduction labor is positive and significant. The same is true for computer share interacted with nonproduction labor.

Table VIII maps these effects into wage changes. In the benchmark interaction specification:

- Narrow outsourcing increases the relative nonproduction wage by 0.507 percentage points per year, while outsourcing outside the two-digit industry decreases it by 0.220. The net outsourcing effect is therefore about 0.287, or roughly 40 percent of the observed increase.
- Computers increase the relative nonproduction wage by 0.557 percentage points per year, or about 75 percent of the observed increase.

This specification gives a larger role to both outsourcing and computers than the linear decomposition.

8.5 Sensitivity to measuring computer capital

The results are most sensitive to the computer measure. Using ex post rental prices gives a relatively large computer effect. Using ex ante rental prices reduces the computer contribution and increases the relative role of outsourcing. Using Census computer investment produces the largest computer effect and a minimal role for outsourcing.

The paper therefore supports the qualitative conclusion that both trade and technology matter, but it cautions against treating the exact numerical split as invariant across measurement choices.

9 How to read the paper's identification strategy

The paper is best read as a model-based accounting and decomposition exercise, not as a randomized or quasi-experimental design. Its identifying logic comes from:

- cross-industry variation in changes in outsourcing and computer capital;
- a theoretical mapping from structural variables to price/productivity changes;
- a second-stage price-regression identity that translates structural components into factor-price components.

The main assumptions are that the structural variables are meaningful proxies for changes in production techniques and that the first-stage decomposition correctly captures the contribution of each structural variable to price and productivity changes.

A limitation is that factor quantities in the interaction specification may be endogenous. The authors acknowledge that they lack strong instruments for factor quantities, so interaction results should be interpreted carefully.

10 Interpretation and economic message

Outsourcing channel. Outsourcing imports the less-skilled-intensive parts of production and leaves relatively skill-intensive tasks in the United States. This raises the relative demand for nonproduction workers.

Technology channel. Computers and high-technology capital complement nonproduction labor and make production more skill-intensive.

Methodological message. Product prices and productivity are endogenous to trade and technology. A price regression that ignores this can be misleading. The paper's solution is to decompose the sources of price/productivity movement before estimating their factor-price implications.

Substantive message. The trade-versus-technology debate should not be framed as either/or. Both foreign outsourcing and computerization contributed to wage inequality in U.S. manufacturing during the 1980s, although the computer contribution tends to be larger in the preferred specifications.

11 Tables and Figures

11.1 Table I: Price Regression, 1979–1990; Table II: Summary Statistics

Read. Table I shows why the conventional price regression is specification-sensitive: with effective TFP, it nearly fits as an identity, while alternative specifications imply different wage changes. Table II reports the underlying data trends: nonproduction wages rose faster than production wages, outsourcing increased, and computer/high-tech capital shares rose during 1979–1990.

TABLE I
DEPENDENT VARIABLE—LOG CHANGE IN INDUSTRY PRICE, 1979–1990

	(1)	(2)	(3)	(4)	(5)
Effective TFP	−1.000	−1.000			
	(0.007)	(0.0006)			
TFP			−0.963	−0.753	
			(0.070)	(0.075)	
Production cost-share	4.680	4.700	3.063	2.428	3.605
	(0.016)	(0.012)	(1.222)	(1.162)	(1.885)
Nonproduction cost-share	5.482	5.443	2.295	4.086	6.202
	(0.019)	(0.031)	(1.430)	(1.722)	(4.036)
Capital cost-share	3.953	3.972	7.888	8.058	9.535
	(0.008)	(0.015)	(0.781)	(0.941)	(2.187)
Materials cost-share times		0.997	1.00*	1.00*	1.219
change in materials price		(0.002)			(0.247)
Energy cost-share times change		0.996	1.00*	1.00*	−0.930
in energy price		(0.006)			(0.915)
constant		0.0101	−0.705	−0.825	−1.929
		(0.005)	(0.301)	(0.293)	(0.915)
R^2	0.999	0.999	0.896	0.806	0.429
N	447	447	447	446	446
					[omitting computers]

Standard errors are in parentheses. All regressions omit three industries with missing data on materials purchases or prices (SIC 2067, 2794, 3483) and are weighted by the industry share of total manufacturing shipments, averaged over the first and last period.

In column (1) the dependent variable is the log change in the industry value-added price and factor cost shares sum to one across primary factors. Effective TFP equals primal TFP minus the change in wage differentials. In columns (2)–(5) the dependent variable is the log change in the gross industry price, and the factor cost shares sum to one across all factors. The materials cost share is multiplied by the log change in the materials price; the energy cost share is treated similarly. Column (2) includes effective TFP as a regressor; column (3) replaces effective TFP with primal TFP; column (4) drops the computer industry (SIC 3573) from the sample; and column (5) also drops the computer industry from the sample and drops TFP as a regressor.

* These coefficients are constrained at unity.

(a) Table I: Dependent variable–log change in industry price, 1979–1990

TABLE II
SUMMARY STATISTICS

	1972–1979		1979–1990	
	Average (percent)	Annual change	Average (percent)	Annual change
Change in log factor prices:				
Production labor		7.666		4.714
Nonproduction labor		7.207		5.437
Capital		8.187		3.954
Materials		9.664		3.485
Energy		15.732		3.250
Factor cost-shares:				
Production labor	12.55	−0.303	10.31	−0.156
Nonproduction labor	6.68	−0.139	6.54	0.002
Capital	24.92	−0.001	27.16	0.262
Materials	50.72	0.170	49.08	−0.261
Energy	2.02	0.113	2.23	−0.023
Other variables:				
TFP (primal)		0.279		0.467
TFP (dual)		0.275		0.467
Outsourcing (broad)	6.31	0.303	9.67	0.363
Outsourcing (narrow)	2.67	0.127	4.40	0.203
Difference	3.64	0.177	5.27	0.160
Using capital services (ex post rental prices):				
High-tech share	7.75	0.125	10.20	0.326
Computer share	4.68	−0.143	5.34	0.198
Difference	3.07	0.267	4.86	0.128
Using capital services (ex ante rental prices):				
High-tech share	5.45	0.115	7.23	0.218
Computer share	3.03	−0.047	3.21	0.053
Difference	2.42	0.162	4.02	0.164
Using computer investment:				
Computer share	1.86	—	3.75	—

Averages are computed over the first and last year of each period (except for the computer investment share which is from 1977 for the 1972–1979 period and the average over 1982 and 1987 for the 1979–1990 period), while changes are measured as an average annual change (the change in log factor prices is the annual average change × 100). Both averages and changes are weighted by the industry share of total manufacturing shipments, except for primary factors, which are weighted by the industry share of total manufacturing payments to that factor. All variables are computed over 447 four-digit SIC industries (excluding SIC 2067, 2794, and 3483), except the High-tech share and Computer share, which are computed over two-digit SIC industries. Those two variables are from the Bureau of Labor Statistics, as used in Berndt, Morrison, and Rosenblum [1992] and Morrison [1997].

Variable definitions:

Outsourcing (broad) = (imported intermediate inputs)/(total nonenergy intermediates) × 100

Outsourcing (narrow) = (imported intermediate inputs in the same two-digit industry as buyer)/(total nonenergy intermediates) × 100

High-tech share = (high-technology capital)/(total capital) × 100

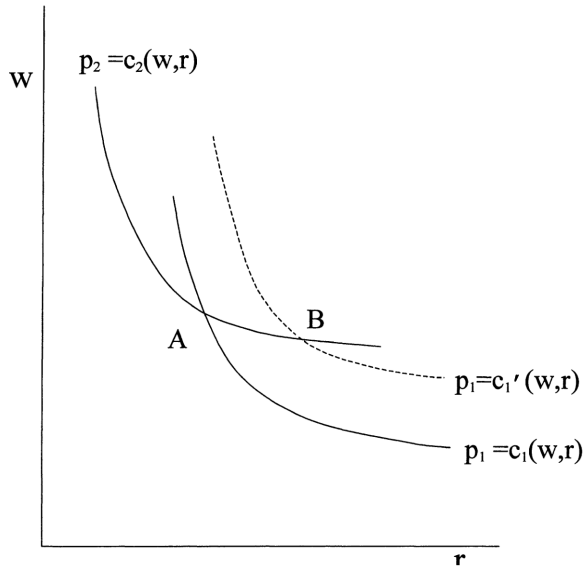
Computer share = (computer equipment)/(total capital) × 100

Computer investment share = (computer investment)/(total investment) × 100

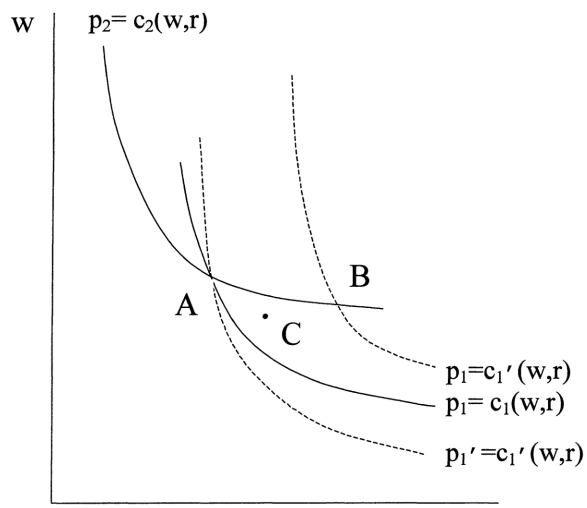
(b) Table II: Summary statistics

11.2 Figure I: Hicks-Neutral Technological Progress; Figure II: Skill-Biased Technological Progress

Read. Figure I shows that neutral technology progress can be offset by product-price adjustment, leaving factor prices unchanged. Figure II shows that skill-biased technological progress changes relative factor demand, so factor prices must adjust even after price feedback.



(a) Figure I: Hicks-neutral technological progress



(b) Figure II: Skill-biased technological progress

11.3 Table III: Nonproduction Wage Share; Table IV: Value-Added Prices plus Effective TFP

Read. Table III gives the direct labor-demand-style benchmark. Table IV is the first-stage linear reduced form, showing how outsourcing and high-technology capital explain the sum of value-added price changes and effective TFP.

TABLE III
DEPENDENT VARIABLE—CHANGE IN NONPRODUCTION WAGE SHARE, 1979–1990

	(1) Mean	(2) Regression	(3) Regression	(4) Regression	(5) Contribution
$\Delta\ln(K/Y)$	0.706	0.042	0.041	0.033	6.0–7.7%
		(0.014)	(0.014)	(0.012)	
$\Delta\ln(Y)$	1.541	0.018	0.016	0.007	2.7–7.1%
		(0.008)	(0.008)	(0.009)	
Outsourcing (narrow)	0.223	0.246	0.265	0.193	11.0–15.2%
		(0.169)	(0.175)	(0.166)	
Outsourcing (difference)	0.200	0.121	0.154	0.038	2.0–7.9%
		(0.046)	(0.050)	(0.054)	
Capital services (ex post rental prices):					
Computer share	0.251	0.206			13.3%
		(0.102)			
High-tech share (difference)	0.144	−0.039			—
		(0.129)			
Capital services (ex ante rental prices):					
Computer share	0.070		0.421		7.6%
			(0.171)		
High-tech share (difference)	0.166		0.014		0.6%
			(0.072)		
Computer investment:					
Computer share	6.561			0.019	31.5%
				(0.007)	
High-tech share (ex post rental prices)	0.395			0.052	5.3%
				(0.051)	
constant		0.207	0.214	0.161	41.5–55.0%
		(0.042)	(0.039)	(0.040)	
R^2		0.163	0.165	0.200	
N		447	447	447	

The mean of the dependent variable equals 0.389. Standard errors (in parentheses) are robust to heteroskedasticity and correlation in the errors within two-digit industries. The first column shows mean values of the dependent and independent variables for 1979–1990. All regressions and means are computed over 447 four-digit SIC industries and are weighted by the average industry share of the manufacturing wage bill. $\Delta\ln(K/Y)$ is the average annual change in the log capital-shipments ratio, and $\Delta\ln(Y)$ is the average annual change in log real shipments. The outsourcing variables and the computer and high-technology shares are in annual changes and are defined in Table II and the text.

(a) Table III: Change in nonproduction wage share, 1979–1990

TABLE IV
DEPENDENT VARIABLE—CHANGE IN VALUE-ADDED PRICES PLUS EFFECTIVE TFP, 1979–1990

	(1)	(2)	(3)
Independent variables:			
Outsourcing (narrow)	0.064	0.080	0.040
	(0.031)	(0.035)	(0.030)
Outsourcing (difference)	0.079	0.113	0.035
	(0.047)	(0.044)	(0.049)
Capital services (ex post rental prices):			
Computer share	0.167		
	(0.066)		
High-tech share (difference)	0.076		
	(0.072)		
Capital services (ex ante rental prices):			
Computer share		0.192	
		(0.108)	
High-tech share (difference)		−0.048	
		(0.082)	
Computer investment:			
Computer share			0.008
			(0.004)
High-tech share (ex post rental prices)			0.093
			(0.049)
constant	4.263	4.294	4.244
	(0.032)	(0.039)	(0.033)
R^2	0.153	0.109	0.213
N	447	447	447

All estimation is over four-digit SIC industries, and equations are weighted by the average industry share of manufacturing shipments. Standard errors (in parentheses) are robust to heteroskedasticity and correlation in the errors within two-digit industry groups. The dependent variable is the log change in value-added prices plus effective TFP (primal total factor productivity minus the change in wage differentials). The outsourcing variables and the computer and high-technology shares are all measured as annual changes; the computer investment share is measured as the average over 1982 and 1987. All variables are defined in

(b) Table IV: Value-added prices plus effective TFP, 1979–1990

11.4 Table V: Estimated Factor-Price Changes; Table VI: Wage Inequality with Alternative High-Tech Measures

Read. Table V is the second-stage linear decomposition using ex post rental prices. Table VI reports the relative nonproduction-production wage effects under alternative measures of computer and high-tech capital.

TABLE V
ESTIMATED FACTOR-PRICE CHANGES—1979–1990

Dependent variable, Change in share-weighted factor prices explained by:	*Outsourcing (narrow)* (1)	*Outsourcing (difference)* (2)	*Computer share* (3)	*High-tech share (difference)* (4)
Mean of dependent variable	0.014	0.013	0.031	0.008
Independent variables:				
Production labor share	−0.010	0.020	−0.005	0.026
	(0.009)	(0.014)	(0.012)	(0.025)
Nonproduction labor share	0.099	0.063	0.248	0.007
	(0.049)	(0.039)	(0.100)	(0.004)
Capital share	0.002	−0.001	0.001	0.004
	(0.003)	(0.003)	(0.004)	(0.004)
R^2	0.256	0.227	0.505	0.310
N	447	447	447	447

Coefficient estimates used to construct the dependent variable are those from column (1) of Table IV. Standard errors are in parentheses and are calculated as described in the text to account for cross-observation correlation in the disturbances that arises from the construction of the dependent variable. Observations are by four-digit SIC industry. All regressions are weighted by the average industry share of total manufacturing shipments.

(a) Table V: Estimated factor-price changes, 1979–1990

TABLE VI
ESTIMATED RISE IN WAGE INEQUALITY—ALTERNATIVE MEASURES OF HIGH-TECHNOLOGY CAPITAL

Dependent variable, Change in value-added prices plus TFP explained by:	*Outsourcing (narrow)*	*Outsourcing (difference)*	*Computer share*	*High-tech share (difference)*
(1) Using BLS capital services (ex post rental prices) for computer share and high-tech share:				
Difference between non-prod. and prod. share	0.108	0.042	0.252	−0.019
	(0.055)	(0.030)	(0.103)	(0.017)
(2) Using BLS capital services (ex ante rental prices) for computer share and high-tech share:				
Difference between non-prod. and prod. share	0.136	0.061	0.143	−0.003
	(0.063)	(0.036)	(0.082)	(n.a.)
(3) Using Census investment flow for computer share and BLS capital services (ex post rental prices) for high-tech share:				
Difference between non-prod. and prod. share	0.069	0.018	0.591	0.118
	(0.051)	(n.a.)	(0.326)	(0.064)

Coefficients shown are the difference between the estimated impact of each dependent variable on the wages of nonproduction labor and the wages of production labor. The row number identifies the column number in Table IV from which coefficient estimates are taken to construct the dependent variable. Standard errors are in parentheses and are calculated as described in the text to account for cross-observation correlation in the disturbances that arises from the construction of the dependent variable. If this method fails to give a positive value for the estimated variance, then "n.a." is reported. Observations are by four-digit SIC industry. All regressions are weighted by the average industry share of total manufacturing shipments.

a. The High-tech share is not measured as a difference from the computer share (i.e., it includes all high-tech capital) when using the *Census* measure of the computer share.

(b) Table VI: Alternative measures of high-technology capital

11.5 Table VII: Interaction First Stage; Table VIII: Interaction Wage-Inequality Decomposition

Read. Table VII estimates the interaction specification, allowing structural variables to interact with factor quantities. Table VIII shows that in the benchmark interaction specification, computers and outsourcing can account for larger shares of the rise in the nonproduction-production wage gap.

TABLE VII
DEPENDENT VARIABLE—CHANGE IN VALUE-ADDED PRICES PLUS EFFECTIVE TFP, 1979–1990

Independent variables:			
		Nonproduction labor interacted with:	
Outsourcing (narrow)	0.666	Outsourcing (narrow)	0.116
	(0.131)		(0.033)
Outsourcing (difference)	−0.248	Outsourcing (difference)	−0.092
	(0.229)		(0.041)
Computer share	0.282	Computer share	0.080
	(0.233)		(0.038)
High-tech share (difference)	1.168	High-tech share (difference)	−0.023
	(0.307)		(0.050)
Production labor interacted with:		*Capital interacted with:*	
Outsourcing (narrow)	−0.002	Outsourcing (narrow)	−0.113
	(0.042)		(0.031)
Outsourcing (difference)	0.156	Outsourcing (difference)	−0.014
	(0.057)		(0.035)
Computer share	0.026	Computer share	−0.054
	(0.047)		(0.041)
High-tech share (difference)	0.302	High-tech share (difference)	−0.291
	(0.094)		(0.084)
Constant	4.259	N	447
	(0.027)	R^2	0.115

The computer share and the high-tech share are measured using BLS capital services (ex post rental prices). All estimation is over four-digit SIC industries, and the regression is weighted by the average industry share of manufacturing shipments. Standard errors (in parentheses) are robust to heteroskedasticity and correlation in the errors within two-digit industry groups. The dependent variable is the log change in value-added prices plus effective TFP (primal total factor productivity minus the change in wage differentials). The outsourcing variables and the computer and high-technology shares are all measured as annual average changes; factor levels are measured as average log quantities for 1979 and 1990.

(a) Table VII: Value-added prices plus effective TFP with interactions

TABLE VIII
ESTIMATED RISE IN WAGE INEQUALITY–ALTERNATIVE MEASURES OF HIGH-TECHNOLOGY CAPITAL

Dependent variable, Change in value-added prices plus TFP explained by:	*Outsourcing (narrow)*	*Outsourcing (difference)*	*Computer share*	*High-tech share (difference)*
(1) Using BLS capital services (ex post rental prices) for computer share and high-tech share:				
Difference between non-prod. and prod. share	0.507	−0.220	0.557	−0.014
	(0.137)	(0.089)	(0.161)	(0.080)
(2) Using BLS capital services (ex ante rental prices) for computer share and high-tech share:				
Difference between non-prod. and prod. share	0.578	−0.018	0.155	0.012
	(0.130)	(0.055)	(0.083)	(0.116)
(3) Using Census investment flow for computer share and BLS capital services (ex post rental prices) for high-tech share:				
Difference between non-prod. and prod. share	0.253	−0.263	0.703	0.169
	(0.169)	(0.106)	(0.288)	(0.248)

Coefficients shown are the difference between the estimated impact of each dependent variable on the wages of nonproduction labor and the wages of production labor. The dependent variables for the regression results reported in row (1) are constructed using coefficient estimates from Table VII. Dependent variables for the regression results reported in rows (2) and (3) are constructed using coefficient estimates from regressions similar to that in Table VII, in which the indicated measures of high-technology capital are used as independent variables. Standard errors are in parentheses and are calculated as described in the text to account for cross-observation correlation in the disturbances that arises from the construction of the dependent variable. Observations are by four-digit SIC industry. All regressions are weighted by the average industry share of total manufacturing shipments.

a. The high-tech share is not measured as a difference from the computer share (i.e., it includes all

(b) Table VIII: Wage inequality under alternative high-tech measures

12 Short summary

Feenstra and Hanson (1999) argue that the conventional price regression, when correctly specified, is an identity and therefore cannot directly identify the sources of wage changes. Their solution is a two-stage procedure: first decompose price and productivity movements into parts attributable to outsourcing and high-technology capital; then use a modified price regression to convert each component into implied factor-price changes.

The main empirical result is that both outsourcing and computers contributed to the rise in the relative wage of nonproduction workers in U.S. manufacturing during 1979–1990. In the base linear specification, computers explain about 35 percent and outsourcing about 15 percent of the relative wage increase. In the interaction specification, the effects can be substantially larger. The exact quantitative split is sensitive to the measure of computer capital, but the qualitative message is that both globalization of intermediate inputs and computerization shifted relative labor demand toward skilled workers.

13 Conclusion

Core conclusion. The rise in U.S. manufacturing wage inequality during the 1980s was not driven solely by trade or solely by technology. Foreign outsourcing and computer-related technological change both raised the relative wage of nonproduction workers.

Methodological takeaway. A fully specified price regression is an accounting identity. To use it for economic explanation, researchers need a first-stage model that assigns price and productivity changes to structural shocks.

Economic intuition. Outsourcing removes relatively less-skilled production stages from domestic manufacturing, while computers complement skilled labor. Both channels increase the domestic relative demand for nonproduction labor.

Important caveat. The numerical magnitude of the computer effect depends heavily on how computer capital is measured. Ex post rental prices, ex ante rental prices, and Census computer investment shares produce different quantitative decompositions.

Useful research lesson. This paper is a strong example of how to connect trade exposure, production structure, technology adoption, product prices, productivity, and factor prices in one empirical framework.